CSE STUDY HUB

DATA COMMUNICATION & NETWORKING UNIT 2

Data Link Layer

SOURCE-ALIGNED CONCISE NOTES

Prepared using the uploaded handwritten class notes for this unit. The explanations retain the concepts and examples visible in those notes, with wording cleaned up for revision.

Topics covered in this unit

- Purpose, framing and MAC addressing
- Transmission time, propagation and delay
- Flow control and Stop-and-Wait ARQ
- Sliding window, Go-Back-N and Selective Repeat

How to use: Learn the definition, draw the diagram/flow where applicable, practise the numerical method, and finish with the revision questions. If a detail is not present in the uploaded pages, it is not added as an asserted class-note topic.

Exam answer structure

For a 5-mark answer: definition → labelled diagram/formula → working or components → example/application → advantage, limitation or comparison.

UNIT 2 · Purpose, framing and MAC addressing

The Data Link Layer provides node-to-node delivery over a single link. The uploaded notes emphasize framing, media access, flow control and error control.

A frame is the Data Link Layer unit. The layer groups a bit stream into identifiable frames so the receiver can detect the beginning and end of each unit.

MAC (Media Access Control) addresses identify network interfaces on a local link. A MAC address is used for local delivery; it is different from an IP address, which supports logical addressing across networks.

Framing approaches include byte count, byte stuffing and bit stuffing. Stuffing inserts extra escape information so data does not accidentally look like a frame delimiter.

UNIT 2 · Transmission time, propagation and delay

Transmission delay is the time needed to place all frame bits onto the link: frame length in bits divided by link bit rate.

Propagation delay is the time taken by a signal to travel through the medium: distance divided by propagation speed. It depends on distance and the medium, not on frame size.

Total one-way delay can include processing delay, queuing delay, transmission delay and propagation delay. Processing delay is the time a device spends examining a packet; queuing delay is time spent waiting in a buffer.

Do not confuse bandwidth with propagation speed. Bandwidth/bit rate determines how quickly bits are put onto the link; propagation speed determines how quickly the signal travels.

UNIT 2 · Flow control and Stop-and-Wait ARQ

Flow control prevents a fast sender from overwhelming a slower receiver. Stop-and-Wait sends one frame and waits for its acknowledgement (ACK) before sending the next frame.

In Stop-and-Wait ARQ, the sender starts a timer after transmitting. If the ACK arrives, it sends the next frame. If the frame or ACK is lost and the timer expires, the sender retransmits.

Sequence numbers help the receiver recognize duplicates. If an ACK is lost, the retransmitted frame may arrive again; the receiver uses the sequence number to avoid delivering duplicate data.

Stop-and-Wait is simple but can waste link capacity on long-delay or high-speed links because the sender remains idle while waiting.

UNIT 2 · Sliding window, Go-Back-N and Selective Repeat

Sliding-window protocols allow multiple frames to be in transit before acknowledgements arrive. The sender window limits how many unacknowledged frames may be outstanding.

Go-Back-N retransmits the lost or damaged frame and the later frames sent after it. The receiver generally accepts frames in order, so several frames may need retransmission.

Selective Repeat retransmits only the specific lost or damaged frames. It is more efficient when errors are occasional, but requires more buffering and more complex receiver logic.

Use Stop-and-Wait for the simplest one-frame-at-a-time explanation; use a sliding-window diagram to show several sequence-numbered frames moving before ACKs return.